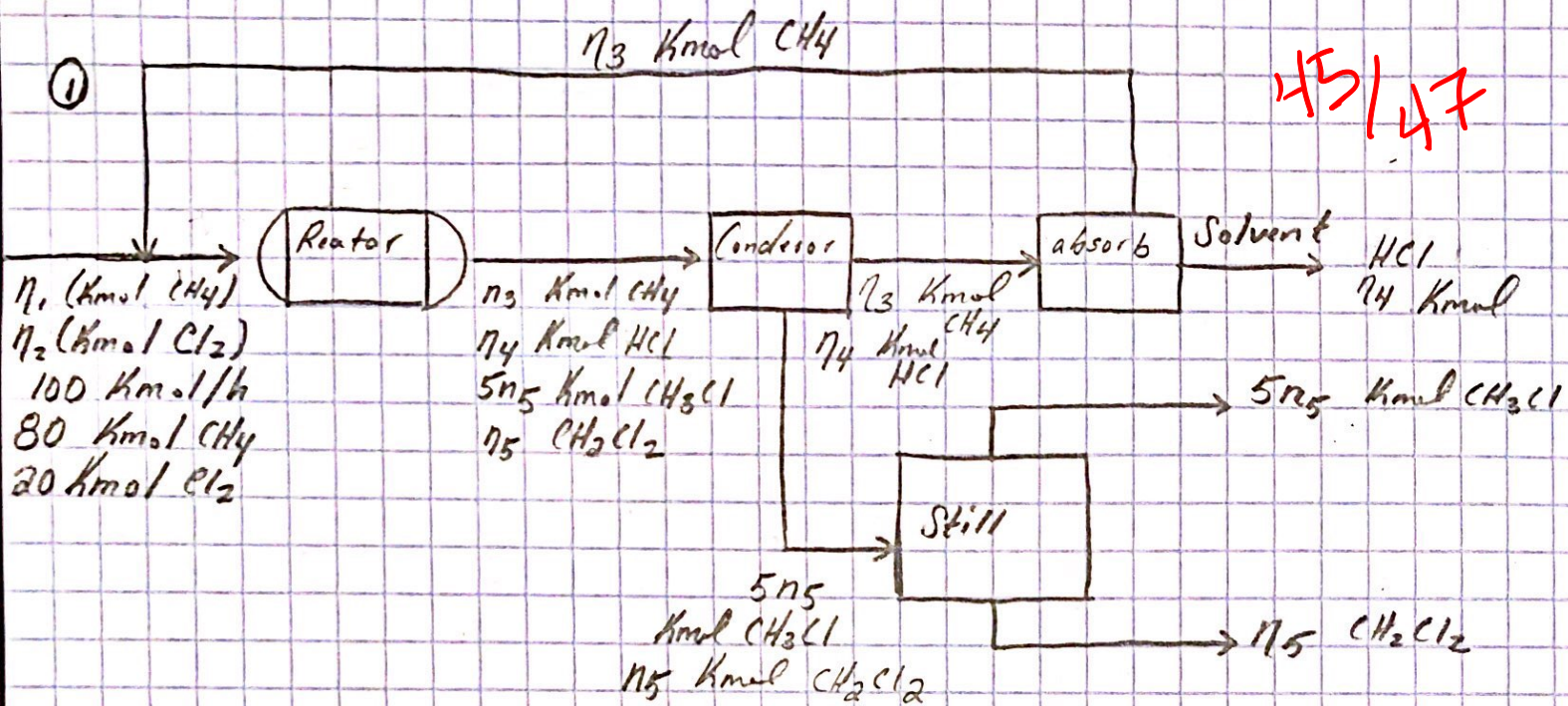


210 hw 8

March 31st

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① Overall DOF: $4 - 3 = 1$

Condensor DOF: $3 - 0 = 3$

Still DOF: $2 - 0 = 2$

Reactor DOF: $3 - 3 = 0$

Absorb DOF: $2 - 0 = 2$

Mixer DOF: $3 - 2 = 1$

Equations:

Cl balance: $2n_2 = n_4 + 5n_5 + 2n_5 \Rightarrow 2n_2 = n_4 + 5n_5 + 2n_5$

C balance: $n_1 = n_3 + 5n_5 + n_5$

H balance: $4 \cdot n_1 = 4n_3 + n_4 + 15n_5 + 2n_5$

① $40 = n_4 + 5n_5 + 2n_5$

② $80 = n_3 + 5n_5 + n_5 \rightarrow n_3 = 80 - 5n_5 - n_5$

③ $320 = 4n_3 + n_4 + 15n_5 + 2n_5$

$320 = 4(80 - 5n_5 - n_5) + n_4 + 15n_5 + 2n_5$

$320 = 320 - 20n_5 - 4n_5 + n_4 + 17n_5$

$0 = 24n_5 - 17n_5 - n_4 \rightarrow n_4 = 24n_5 - 17n_5 \rightarrow n_4 = 7n_5$

① $40 = 7n_5 + 5n_5 + 2n_5 \rightarrow 40 = 14n_5 \rightarrow n_5 = 2.86 \text{ kmol } \text{CH}_2\text{Cl}_2/\text{h}$

$5n_5 \rightarrow 5(2.86) = 14.3 \text{ kmol } \text{CH}_3\text{Cl}/\text{h}$

$n_4 = 7(2.86) \rightarrow n_4 = 20.02 \text{ kmol } \text{HCl}/\text{h}$

Mixer equations

$$\eta_1 + \eta_3 = 80, \quad \eta_2 = 20$$

$$\text{eg } (2): \eta_3 = 80 - 5\eta_5 - \eta_5 \rightarrow \eta_3 = 80 - 14.3 - 2.86 = 62.86 \frac{\text{kmol CH}_4}{\text{h}}$$

$$\eta_1 + \eta_3 = 80 \rightarrow \eta_1 = 17.14 \text{ kmol CH}_4/\text{h}$$

$$\boxed{\begin{aligned} \eta_1 &= 17.14 \text{ kmol CH}_4/\text{h}, \quad \eta_2 = 20 \text{ kmol Cl}_2/\text{h}, \quad \eta_3 = 62.86 \text{ kmol CH}_4/\text{h} \\ \eta_4 &= 20.02 \text{ kmol HCl}/\text{h}, \quad \eta_5 = 2.86 \text{ CH}_2\text{Cl}_2/\text{h} \end{aligned}}$$

c. $1000 \text{ kg CH}_3\text{Cl}/\text{h}$ $\text{CH}_3\text{Cl} = 50.48 \text{ kg CH}_3\text{Cl}/\text{kmol}$

$$\frac{1000 \text{ kg CH}_3\text{Cl}}{\text{h}} \times \frac{1 \text{ kmol CH}_3\text{Cl}}{50.48 \text{ kg CH}_3\text{Cl}} = \frac{19.81 \text{ kmol CH}_3\text{Cl}}{\text{h}}$$

$$\text{Scale factor} = \frac{19.81 \frac{\text{kmol CH}_3\text{Cl}}{\text{h}}}{14.29 \frac{\text{kmol CH}_3\text{Cl}}{\text{h}}} = 1.39$$

Feed Stream with

Scale factor :

$$\begin{aligned} \eta_1 &= 17.14 \cdot 1.39 = 23.82 \text{ kmol CH}_4/\text{h} \\ \eta_2 &= 20 \cdot 1.39 = 27.80 \text{ kmol Cl}_2/\text{h} \end{aligned} \quad \left. \begin{array}{l} \text{feed Stream} = \\ 51.62 \text{ kmol} \\ \text{fresh feed} \end{array} \right\}$$

Recycle Stream:

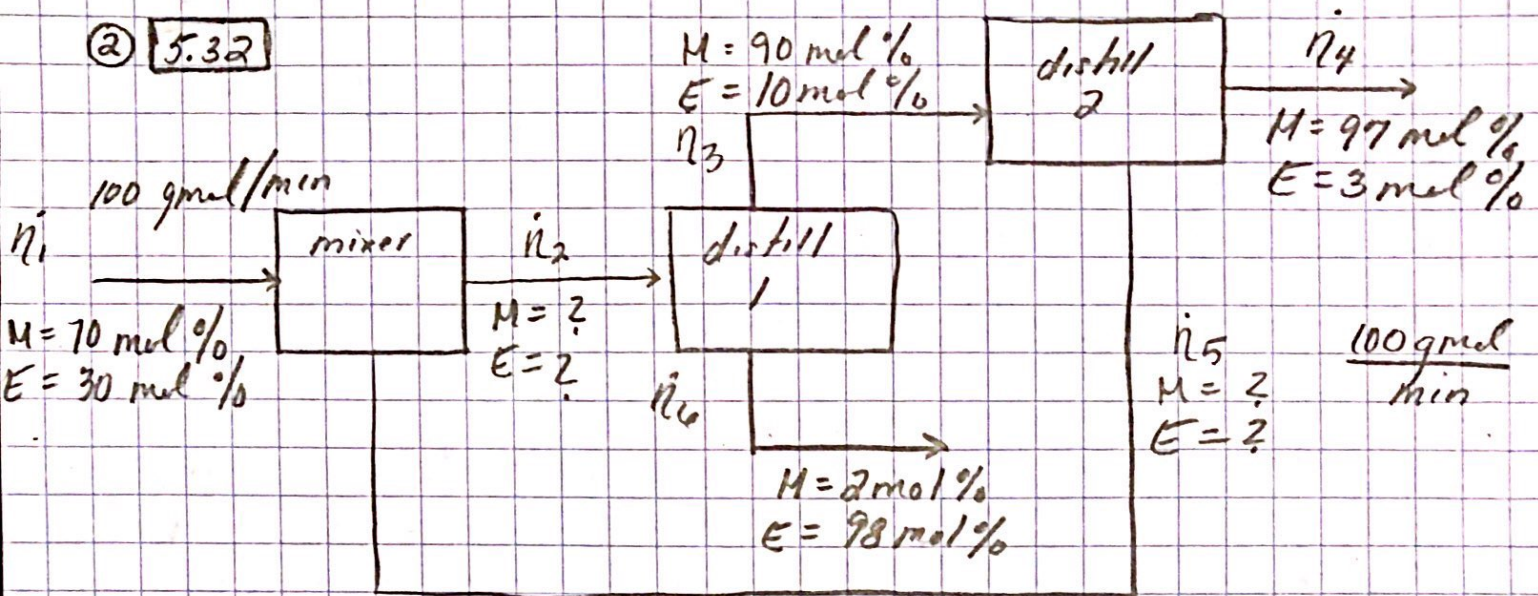
$$\eta_3 = 62.86 \cdot 1.39 = 87.38 \text{ kmol CH}_4/\text{h}$$

Compositions:

$$\text{CH}_4: \frac{23.82 \text{ kmol CH}_4/\text{h}}{51.62 \text{ kmol feed}} = .46 \rightarrow 46\% \text{ CH}_4$$

$$\text{Cl}_2: \frac{27.80 \text{ kmol Cl}_2/\text{h}}{51.62 \text{ kmol feed}} = .539 \rightarrow 53.9\% \text{ Cl}_2$$

② 5.32



$$\text{DOF: } (8 - 8) = 0$$

Mixer equations:

$$40 + X_{M5}(100) = X_{M5}\dot{n}_2$$

$$30 + X_{E5}(100) = X_{E5}\dot{n}_2$$

$$X_{M5}\dot{n}_2 = 0.9\dot{n}_3 + 0.02\dot{n}_6 \rightarrow (1 - X_{E2})\dot{n}_2 = 9\dot{n}_3 + 0.02\dot{n}_6$$

Distillation 1:

$$0.90\dot{n}_3 = 0.97\dot{n}_4 + X_{M5}(100)$$

$$X_{E2}\dot{n}_2 = 0.1\dot{n}_3 + 0.98\dot{n}_6$$

Distillation 2:

$$0.1\dot{n}_3 = 0.03\dot{n}_4 + X_{E5}(100) \rightarrow X_{E5}(100) = 0.1\dot{n}_3 - 0.03\dot{n}_4$$

$$30 + 0.1\dot{n}_3 - 0.03\dot{n}_4 = 0.1\dot{n}_3 + 0.98\dot{n}_6$$

$$0.90\dot{n}_3 = 0.97\dot{n}_4 + X_{M5}(100) \rightarrow X_{M5}(100) = 0.9\dot{n}_3 - 0.97\dot{n}_4$$

$$70 + 0.9\dot{n}_3 - 0.97\dot{n}_4 = X_{M2}\dot{n}_2$$

$$70 + 0.9\dot{n}_3 - 0.97\dot{n}_4 = 0.9\dot{n}_3 + 0.02\dot{n}_6$$

$$70 = 0.97\dot{n}_4 + 0.02\dot{n}_6$$

$$30 = .03 \dot{n}_4 + 0.98 \dot{n}_6$$

$$70 = 0.97 \dot{n}_4 + 0.02 \dot{n}_6$$

$$30 - .03 \dot{n}_4 = 0.98 \dot{n}_6 \quad \dot{n}_6 = \frac{30 - .03 \dot{n}_4}{.98}$$

$$70 = 0.97 \dot{n}_4 + .02(30.6 - .0306 \dot{n}_4)$$

$$70 = 0.97 \dot{n}_4 + 0.612 - .000612 \dot{n}_4$$

$$69.41 = 0.97 \dot{n}_4 \rightarrow \dot{n}_4 = 71.5 \text{ g mol/min}$$

$$\dot{n}_6 = \frac{30 - .03(71.5)}{.98} = 28.42 \text{ g mol/min}$$

$$\dot{n}_3 = \dot{n}_4 + \dot{n}_6 \rightarrow \dot{n}_3 = 71.58 + 100 \rightarrow 171.58 \text{ g mol/min} = \dot{n}_3$$

$$0.1 \dot{n}_3 = .03 \dot{n}_4 + x_{E5}(100) \rightarrow 0.1(171.58) = .03(71.5) + x_{E5}(100)$$

$$17.16 = 2.145 + x_{E5}(100) \rightarrow x_{E5} = .15 \quad x_{M5} = 1 - .15 = .85$$

$$x_{M2} \dot{n}_2 = 0.9 \dot{n}_3 + .02 \dot{n}_6 \rightarrow x_{M2} \dot{n}_2 = .9(171.58) + .02(28.4) = 154.99 \frac{\text{g mol}}{\text{min}}$$

$$x_{E2} \dot{n}_2 = 0.1 \dot{n}_3 + 0.98 \dot{n}_6 \Rightarrow x_{E2} \dot{n}_2 = 45 \text{ g mol/min}$$

$$x_{M2} = 1 - x_{E2} \rightarrow \dot{n}_2 = \frac{154.99}{1 - x_{E2}} \quad \dot{n}_2 = \frac{45}{x_{E2}} \rightarrow \frac{45}{.225} = 199.9 \frac{\text{g mol}}{\text{min}}$$

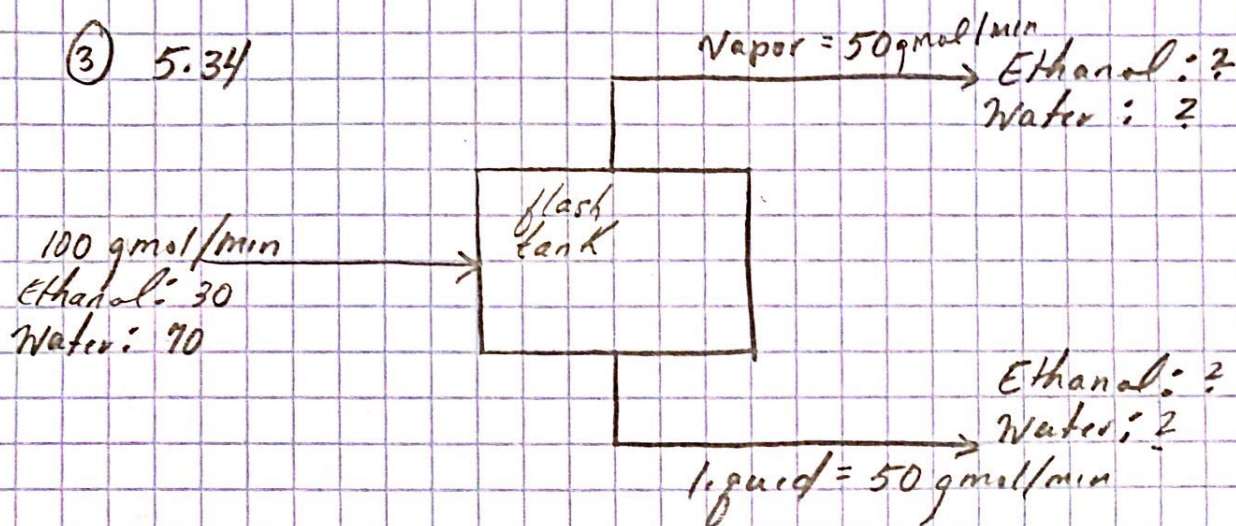
$$154.99 x_{E2} = 45 - 45 x_{E2} \rightarrow x_{E2} = 0.225$$

$$x_{M2} = 1 - 0.225 = 0.775$$

$$f \text{ for Methane: } \frac{(.97)(71.58)}{70} = .992 \rightarrow 99.2\%$$

$$f \text{ for Ethanol: } \frac{(.98)(28.42)}{30} = .928 \rightarrow 92.8\%$$

③ 5.34



$$\begin{aligned} 0.3(100) &= 30 \rightarrow 30 = L_E(50) + V_E(50) \\ 0.7(100) &= 70 \rightarrow 70 = L_W(50) + V_W(50) \\ 1 &= V_E + V_W \\ 1 &= L_W + L_E \end{aligned}$$

Data from table 13.11

$$\begin{aligned} @ 84.1^\circ\text{C} \quad L_E &= 0.1661 \quad V_E = 0.5089 \\ @ 85.3^\circ\text{C} \quad L_E &= 0.1238 \quad V_E = 0.4704 \end{aligned}$$

$$@ 84.1^\circ\text{C}: (50)(0.1661) + (50)(0.5089) = 33.75$$

$$@ 85.3^\circ\text{C}: (50)(0.1238) + (50)(0.4704) = 29.71 \rightarrow \text{close to } 30 \checkmark$$

85.3°C

Fractional Recoveries

$$\text{Ethanol: } \frac{(0.47)(50)}{(0.3)(100)} = .78 \quad \text{for Water: } 1 - 0.1238 = .8762$$

$$\text{Water: } \frac{(.8762)(50)}{(.7)(100)} = .6259$$

$$\text{Separation factor: } \frac{(0.4704)(.8762)}{(0.1238)(.6259)} = 5.319$$



Comment Summary

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1. -2

Check the denominator.